MIDTERM REPORT

Richard Xu

Github repo: https://github.com/Richard098789/JSC370_Final_Project

1. Introduction

The rapid expansion of data-driven industries has increased demand for professionals in roles such as data analysts, data scientists, and data engineers. Online job boards aggregate thousands of postings, but advertised compensation often varies widely and is incompletely reported: listings may omit salary, report ranges in different units (hourly versus annual), or describe location at anything from city to country level. That heterogeneity makes it hard for candidates and researchers to interpret typical pay from raw postings alone.

This project focuses on a narrower, testable question than a broad labour-market survey:

Question A . Among Canadian data-related postings that disclose salary, which job and market characteristics are most associated with advertised pay—especially role family, seniority, contract type, geography, labour-market demand, and regional cost of living proxied by CPI?

Question B. How well can machine learning models predict advertised salary on a held-out test set from those characteristics, and what do model comparisons and diagnostics suggest about nonlinearity and variable importance?

Data in brief. Postings come from the Adzuna Jobs API for Canada with a "data" keyword query. The analytic frame requires non-missing salary minima, applies title-based role rules, standardizes pay, maps locations to provinces (with Canada-unspecified when only country-level text is available), merges external CPI from Statistique Canada and restricts to a post-April 2025 window for consistency.

2. Methods

2.1 Data source and scope

Primary data. The Adzuna API aggregates listings from multiple sources (employers, agencies, boards) and returns structured fields including title, company, location hierarchy, salary min and max, contract type, category, and posting time. This project queries the Canadian endpoint with what="data" and paginates results (up to about 5,000 raw rows when 100 pages times 50 results are requested; the exact row count depends on API responses and de-duplication by job id).

External data. Consumer Price Index values, used as a regional cost-of-living proxy, were compiled for Canada, provinces, and selected cities and merged to postings.

Limitations. Selection into salary disclosure: postings without salary cannot support the salary-focused models and are dropped for those analyses; conclusions may tilt toward employers who publish ranges. API coverage: Adzuna is not a census of all Canadian vacancies. Location: Canada-unspecified denotes listings only to country-level Canada under the rule set, including most of the **remote job postings**; these are kept with an explicit indicator for modeling rather than guessing a province.

2.2 Data cleaning and wrangling

Salary. Minimum and maximum advertised amounts are combined into a midpoint; values that appear to be hourly, daily, weekly, or monthly are converted to annual equivalents using fixed heuristics (thresholds on magnitude), then log-transformed using log1p for modeling to stabilize skewness and variance. Rows with missing salary_min are excluded from salary-focused modeling.

Location and geography. Raw location strings are normalized; a province label is assigned using keyword lists (cities and region names). A major-city flag uses the finer location field when available. Canada-unspecified is used for country-only labels.

Time window. For the main tables and supervised models, the project uses postings with created on or after 2025-04-01.

Tools. Data assembly and cleaning use Python (pandas, numpy, requests); visualization uses matplotlib and seaborn and Plotly; modeling uses scikit-learn, XGBoost, and pyGAM.

2.3 Feature engineering

1) Role family. Titles are cleaned (lowercase, punctuation removal) and mapped to categories (for example Data Scientist, Data Engineer, Data Analyst, Manager, Data Entry, Developer, Other) to reduce noise from near-duplicate strings.

2) Demand proxy. Within-sample role frequency is used as a coarse market-thickness measure (role_demand, log-scaled).

3) Seniority. is_senior flags senior, lead, or principal-type wording.

4) Contract. is_permanent maps permanent versus contract when available.

5) Cost of living. cpi from merged regional CPI.

6) Time. year and month from posting date were included to capture calendar trend and within-year seasonality.

7) Modeling encoding for tree, ensemble, and GAM block. role_encoded is target-encoded using the training split only (median salary by role on the original scale, then log1p), with a fallback median for rare roles. Test rows never define the encoding (avoids leakage).

8) Missing province signal. missing_province indicates Canada-unspecified locations.

9) Correlation. Before fitting supervised models, pairwise Pearson correlations among predictors on the training split are examined (heatmap plus sorted pairs). In this project, year and month can be strongly correlated (magnitude of r often near 0.9 or more in the training data) because the two fields jointly trace posting date, so they carry redundant linear information. Random forests and gradient boosted trees can still use both for prediction. The report does not arbitrarily drop one of year or month here; instead we acknowledge the redundancy and focus test metrics on out-of-sample predictive accuracy.

2.4 Exploratory data analysis (EDA)

EDA uses distributional plots (histograms of salary), comparative plots (boxplots by role, contract type, province), counts (postings by role), and time series of posting volume and mean or median salary by date or month. Interactive Plotly versions of several displays are used on the website for hover detail (title, company, location, pay).

2.5 Supervised models, validation, and test metrics

Outcome and split. The regression target is log1p(salary_mid). I use train_test_split with test_size=0.2 and random_state=42 on the modeling frame after filters. All compared models use the same X_train, X_test, y_train, y_test.

Predictors. role_encoded, is_senior, is_permanent, is_major_city, cpi, role_demand, year, month, missing_province.

Metrics. RMSE and MAE are reported on the original salary (dollar) scale by back-transforming predictions with expm1 after log1p modeling. R-squared is computed on that same scale. Cross-validated RMSE for random forest and XGBoost uses a custom scorer that inverts the log transform inside each fold so CV matches dollars.

Baseline. Predict every test observation with the training mean of log1p(salary), then transform back—a reference for how much RMSE improves with richer models.

Random Forest. RandomForestRegressor with n_estimators=200, max_depth=None, random_state=42, n_jobs=-1. Five-fold CV on the training set for RMSE on the original scale, then refit on full training for test evaluation.

Generalized Additive Model (pyGAM). LinearGAM with spline terms s() on role_encoded, cpi, role_demand, year, month, and linear terms l() on binary predictors. Five-fold CV fits a median SimpleImputer on training folds only (pyGAM requires finite X); the final model refits the imputer on all training rows before test prediction.

XGBoost. XGBRegressor with objective reg:squarederror, subsample=0.8, colsample_bytree=0.8, min_child_weight=1, gamma=0, random_state=42. GridSearchCV with five-fold CV over n_estimators in {200, 300}, learning_rate in {0.025, 0.05, 0.1}, max_depth in {2, 4}, optimizing the same original-scale RMSE scorer. The best estimator is evaluated on the held-out test set.

Additional interpretability. XGBoost feature importances and partial dependence (for example on year and month). **A separate ordinary least squares model** with one-hot role dummies on role; it complements but is not identical to the main test-metric pipeline.

3. Results

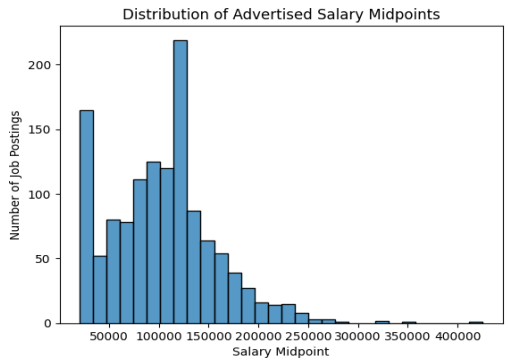

Figure 1. Histogram of salary

Figure 2. Salary by role (interactive)

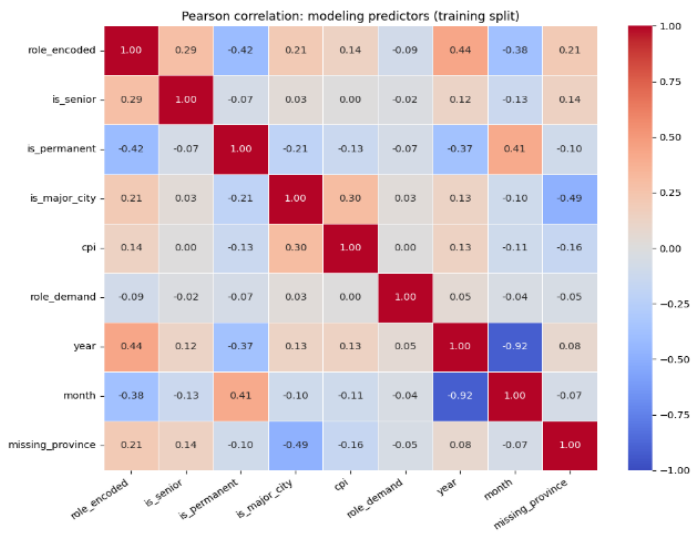

Figure 6. Feature correlation heatmap

Analytic sample summary

Canadian data-related postings; Methods define date restriction and salary standardization.

Statistic	Value
Sample size (number of postings, N)	1,285
Distinct job-posting IDs	1,285
First posting date	2025-04-12
Last posting date	2026-04-26
Calendar span	378 days
Distinct role categories (classified)	7
Distinct province / region labels	6
Postings with Canada-unspecified province	275 (21.4%)
Median advertised salary midpoint	$105,000 CAD
Interquartile range — salary midpoint	$67,500 – $130,000 CAD

Table 1. Modeling sample summary

Model Performance Comparison

XGBoost achieved the lowest validation and test error.

Model	CV RMSE	Test RMSE	Test MAE	Test R^2
Baseline	—	$55,916	$43,253	-0.107
GAM	$39,713	$40,000	$25,907	0.433
Random Forest	$39,289	$42,782	$27,155	0.352
XGBoost	$37,900	$39,749	$25,105	0.441

Table 2. Model comparison statistic

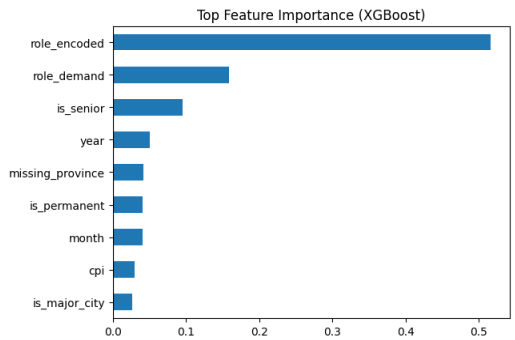

Figure 7. Feature importance overall

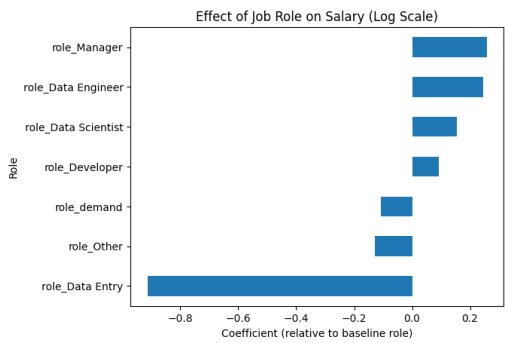

Figure 8. Feature importance on each role

Figure 9. XGBoost Actual vs Predicted Salary (interactive)

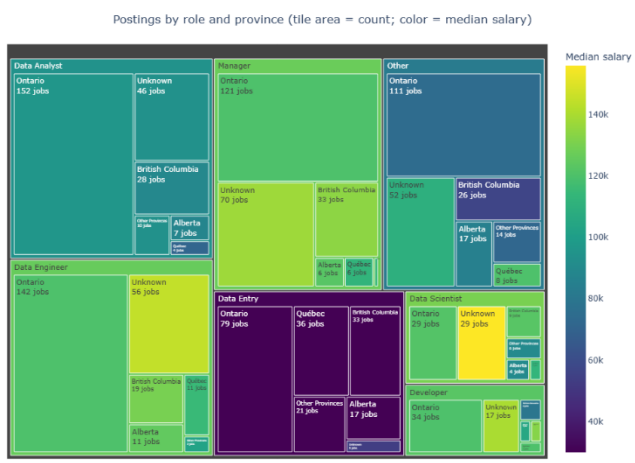

Figure 10. Postings by role and province (interactive)

XGBoost achieves the strongest overall performance among the models evaluated. Its **test RMSE is $39,749**, which is the lowest across all models and represents a substantial improvement over the baseline RMSE of **$55,916** (a reduction of approximately **$16,167**, or about **29%**). Compared to Random Forest (**$42,782 RMSE**), XGBoost reduces error by about **$3,033 (~7%)**, and compared to GAM (**$40,000 RMSE**), the improvement is relatively modest at **$251 (~0.6%)**.

The **R^2 value for XGBoost is 0.441**, meaning the model explains approximately **44.1% of the variance** in salary. This is slightly higher than GAM (**0.433**) and notably higher than Random Forest (**0.352**). However, the relatively moderate R^2 across all models reflects the inherent noise and variability in advertised salary data.

4. Conclusion

The results consistently show that **job role is the primary driver of salary variation**, with role-related features accounting for the majority of predictive power. Roles such as Data Engineer and Manager are associated with substantially higher salaries, while Data Entry positions occupy the lower end of the distribution. This pattern reflects the differing levels of technical specialization and responsibility across job categories. In addition, **seniority plays an important role**, as positions labeled “Senior” or “Lead” tend to command higher salaries, consistent with expectations regarding experience-based compensation.

Beyond role and seniority, **market demand contributes meaningfully to salary differences**. Roles that appear more frequently in the dataset—used here as a proxy for demand—tend to be associated with higher salaries, suggesting that employers may offer competitive compensation in response to labor market conditions. In contrast, **geographic variables such as cost of living (CPI) and major city indicators have a more modest impact**. While there is some evidence that higher-cost regions correspond to higher salaries, these effects are relatively small compared to role-based differences. This suggests that, within the context of job postings, compensation is driven more by the nature of the work than by location alone.

From a predictive perspective, **XGBoost provides the best overall performance**, achieving the lowest test RMSE and highest R^2 among the models considered. However, the improvements over Random Forest and the Generalized Additive Model (GAM) are moderate rather than substantial, indicating that much of the structure in the data can be captured by simpler or more interpretable approaches. The relatively modest R^2 values across all models highlight the inherent variability and noise in advertised salary data, where differences in reporting practices, incomplete information, and heterogeneous compensation structures limit the achievable level of prediction.

Several limitations should be considered when interpreting these findings. First, the analysis relies on job postings obtained through the Adzuna API, which does not represent a complete or unbiased sample of the labor market. In particular, **salary information is not uniformly disclosed**, and postings without salary data were excluded from the modeling stage. This introduces a potential selection bias toward employers who publish compensation ranges. Second, **location information is often incomplete or imprecise**, with some postings labeled only at the country level ("Canada-unspecified"), limiting the ability to fully capture geographic variation.

From a modeling perspective, techniques such as target encoding and tree-based methods may fit patterns in relatively small samples, raising the possibility of overfitting despite cross-validation. Additionally, certain predictors—such as year and month—are highly correlated in the dataset, making it difficult to interpret their independent effects. Finally, the relationships identified in this study are associational. The analysis is based on observed job postings, and therefore cannot establish whether specific factors directly determine salary outcomes.

Future work could address these limitations by incorporating **external validation data**, such as official wage statistics, and by expanding the dataset to include more observations over a longer time horizon. More precise geographic data, including city-level economic indicators, would improve the measurement of location effects. Additionally, alternative modeling approaches, such as time-series validation or hierarchical models, could provide further insight into how salary dynamics evolve over time and across regions.

In summary, this study demonstrates that salary prediction for data-related jobs can be effectively approached using machine learning techniques, with role, demand, and seniority emerging as the dominant drivers. While geographic and economic factors contribute to variation, their influence is secondary. These findings provide a clearer understanding of how compensation is structured in the data job market and offer a foundation for further exploration using richer and more comprehensive data sources.